

Single-Image Super-Resolution with SinGAN: An Empirical Study of Noise, Scale, and Capacity

1 Problem Definition

Modern deep learning models typically require large-scale datasets, but collecting such data is often expensive or impractical. This project investigates whether meaningful single-image super-resolution (SR) can be achieved using only one photograph. Given a low-resolution image $I_{LR} \in \mathbb{R}^{\frac{H}{s} \times \frac{W}{s} \times 3}$, the goal is to recover a high-resolution image $\hat{I}_{HR} \in \mathbb{R}^{H \times W \times 3}$ such that $\hat{I}_{HR} \approx I_{HR}$, where s is the upscaling factor. We apply SinGAN [3], a multi-scale generative adversarial network that learns internal patch statistics from a single image, to the JCSMR building photograph (5616×3744 px). We systematically study the effect of three key parameters (noise_amp, scale_factor, and nfc) on SR quality. Performance is measured using PSNR and SSIM as distortion metrics, and BRISQUE as a no-reference perceptual quality metric, following the observation that distortion and perceptual metrics capture complementary aspects of SR quality [1]. We consider the approach successful if any SinGAN configuration exceeds the bicubic interpolation baseline (PSNR=28.66 dB, SSIM=0.8932, BRISQUE=38.08).

2 Method

2.1 SinGAN Architecture

SinGAN [3] learns a generative model from a single image by training a multi-scale pyramid of patch-GANs. Each scale n consists of a generator G_n and a discriminator D_n . The generator produces an image at scale n by upsampling the output of the previous scale and adding learned noise, while the discriminator distinguishes between real patches from the training image and generated patches at that scale. Formally, at each scale the generator minimises:

$$\mathcal{L}(G_n, D_n) = \mathcal{L}_{\text{adv}}(G_n, D_n) + \alpha \mathcal{L}_{\text{rec}}(G_n) \quad (1)$$

where $\mathcal{L}_{\text{adv}}$ is the WGAN-GP adversarial loss [3] and $\mathcal{L}_{\text{rec}}$ is a reconstruction loss weighted by α . For super-resolution, the coarsest scale is initialised with the low-resolution input, and each successive scale refines the output. The per-scale upsampling ratio r is determined by:

$$r = s^{1/k} \quad (2)$$

where s is the total upscaling factor and k is the number of pyramid levels, itself determined by `scale_factor`.

2.2 Experimental Setup

The input image (5616×3744 px) was downsampled to 256×256 to reduce computational cost while preserving structural patterns. We set `sr_factor`=2 throughout, as our goal is to study parameter sensitivity rather than maximise upscaling. All experiments use 500 iterations per scale to enable rapid controlled comparison across configurations; we treat this as a fixed constant so that observed differences in quality are attributable to the varied parameter alone. A bicubic upsampling baseline is included as a reference, as it represents the standard non-learned comparison point in SR literature.

A limitation of this setup is that 500 iterations per scale may underfit relative to the default 2000, potentially underestimating the quality ceiling of each configuration.

The full pipeline is illustrated in Figure 1.

2.3 Ablation Study

We conduct a controlled ablation varying one parameter at a time while holding all others fixed at the baseline values ($\text{noise_amp}=0.10$, $\text{scale_factor}=0.85$, $\text{nfc}=32$, $\text{sr_factor}=2$); default SinGAN parameters are optimised for general generation rather than architectural SR, motivating this tuning study (Figure 2). Three parameters are studied: (1) $\text{noise_amp} \in \{0.05, 0.10, 0.15\}$, controlling noise magnitude and thus sharpness; (2) $\text{scale_factor} \in \{0.80, 0.85, 0.90\}$, determining pyramid depth k ; and (3) $\text{nfc} \in \{32, 64\}$, setting convolutional channel capacity.

2.4 Evaluation Protocol

The original image is downsampled by a factor of $s = 2$ (bicubic) to obtain I_{LR} . Each model reconstructs $\hat{I}_{HR}$, which is compared against I_{HR} . Metrics are computed using `scikit-image` (PSNR, SSIM) and `piq` (BRISQUE)..

3 Results

Table 1 presents quantitative results across all configurations. E4-B ($\text{nfc}=64$) achieves the best distortion performance (PSNR=34.09 dB, SSIM=0.9729), surpassing the bicubic baseline by 5.43 dB, confirming that representational capacity is the dominant factor in SR quality. The noise_amp ablation reveals a threshold effect: only $\text{noise_amp}=0.15$ exceeds bicubic on PSNR (29.47 dB), suggesting a minimum noise level is required to synthesise meaningful high-frequency detail. The scale_factor ablation produces non-monotonic results, with E3-C ($\text{scale}=0.90$) introducing visible colour as in Figure 3(f) consistent with error accumulation across pyramid levels.

A key finding is the disagreement between metrics: E3-C achieves SSIM=0.9553 despite visible colour drift, confirming SSIM is insensitive to colour shifts, while BRISQUE correctly penalises it (22.91). Furthermore, all SinGAN configurations achieve substantially lower BRISQUE than bicubic (38.08), demonstrating that SinGAN produces more perceptually natural outputs than interpolation even when pixel accuracy is lower. This distortion, perception tradeoff [1] confirms that no single metric sufficiently characterises SR quality.

4 Reflection

Single-image SR has broad societal implications such as satellite imaging, historical preservation, and medical diagnostics, where high-resolution imagery is costly to acquire directly. However, Generative methods such as SinGAN *hallucinate* detail rather than recover it, which is concerning in forensic contexts: invented texture in surveillance footage could be mistaken for real evidence, potentially influencing legal outcomes incorrectly [2]. Additionally, SR can enhance footage sufficiently to identify individuals who would otherwise remain anonymous, raising serious privacy concerns. Both risks are compounded by the architectural bias of our model, which may fail silently on faces or other scene types.

5 Conclusion

This project demonstrates that SinGAN achieves effective single-image super-resolution, with most configurations outperforming bicubic interpolation. Channel capacity emerged as the dominant factor, while sufficient noise was required for high-frequency detail synthesis. Metric disagreement highlighted the need for complementary evaluation. Future work includes incorporating perceptual loss, improving generalisation, and increasing training iterations.

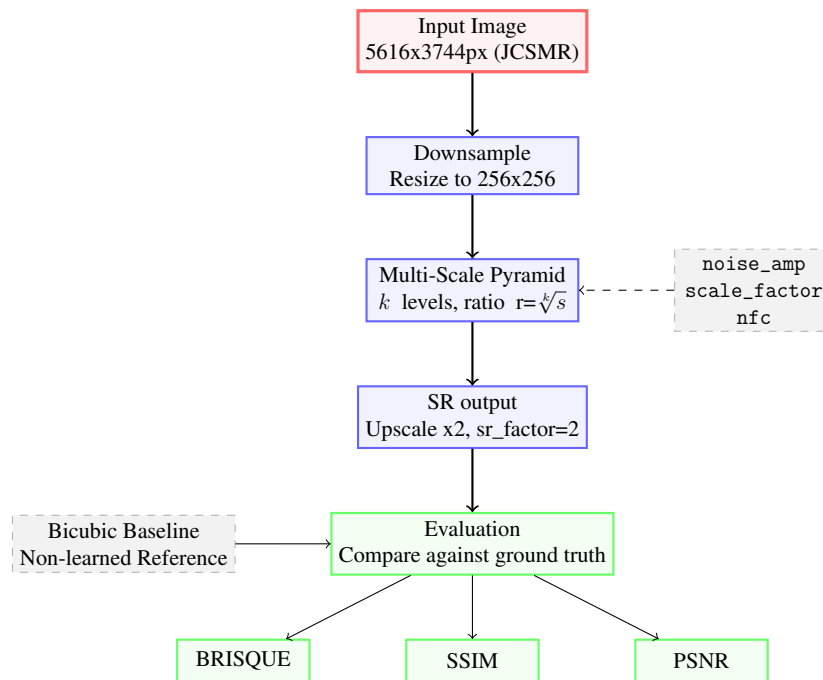


Figure 1: Overview of the proposed SR pipeline. The input image is downsampled and used to train a multi-scale SinGAN pyramid. Three hyperparameters (`noise_amp`, `scale_factor`, `nfc`) are varied in a controlled ablation (dashed). SR outputs are evaluated against the original using PSNR, SSIM, and BRISQUE, with bicubic interpolation as a non-learned baseline.

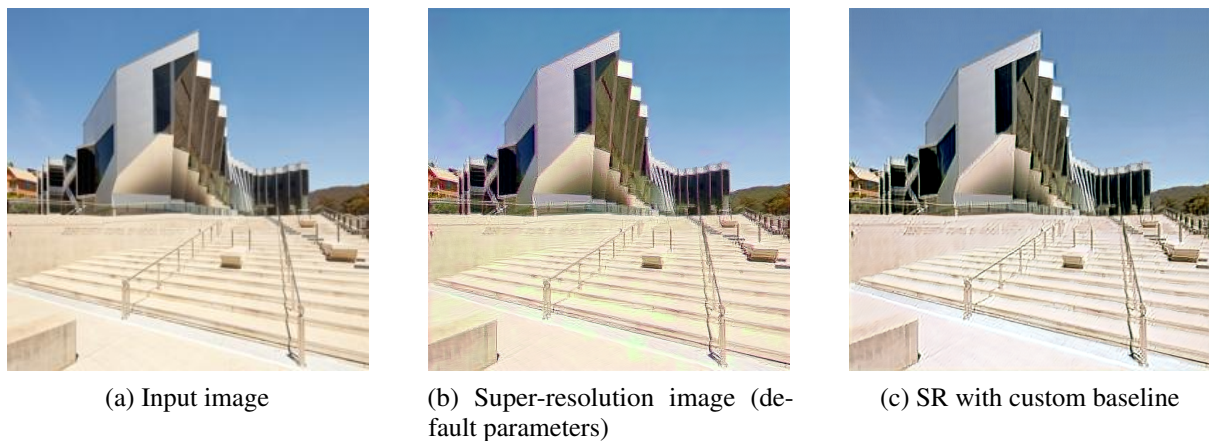


Figure 2: Comparison of super-resolution outputs. (a) Input low-resolution image. (b) Default SinGAN SR output. (c) Custom baseline (`noise_amp`=0.10, `scale_factor`=0.85, `nfc`=32).



(a) Original Image



(b) E2-A: noise=0.05



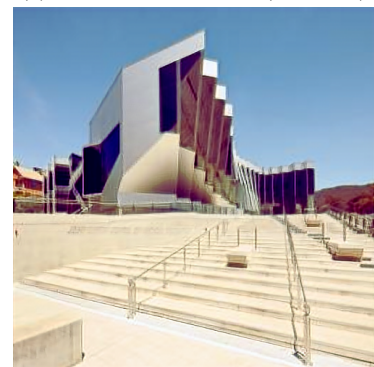
(c) E2-B: noise=0.10 (baseline)



(d) E2-C: noise=0.15



(e) E3-A: scale=0.80



(f) E3-C: scale=0.90 (colour drift visible)



(g) E4-B: nfc=64 (best PSNR)

Figure 3: Qualitative comparison of SinGAN SR outputs across parameter configurations. (a) Original input image. (b)–(d) vary noise_amp (0.05, 0.10, 0.15); (e)–(f) vary scale_factor (0.80, 0.90); (g) doubles channel capacity (nfc=64). Note the visible purple colour drift in (f) (scale_factor=0.90) and the sharper facade detail in (g) (nfc=64).

Table 1: Quantitative evaluation of SinGAN parameter configurations for single-image super-resolution. PSNR and SSIM are distortion metrics (higher is better); BRISQUE is a no-reference perceptual quality metric (lower is better). Bold denotes best performance per metric. Baseline values: noise_amp=0.10, scale_factor=0.85, nfc=32, sr_factor=2.

Method	PSNR (dB)	SSIM	BRISQUE
Bicubic baseline	28.66	0.8932	38.08
E2-A noise=0.05	27.08	0.8925	18.13
E2-B noise=0.10	27.13	0.9349	17.78
E2-C noise=0.15	29.47	0.9356	16.63
E3-A scale=0.80	30.54	0.9629	19.36
E3-B scale=0.85	27.13	0.9349	17.78
E3-C scale=0.90	28.89	0.9553	22.91
E4-A nfc=32	27.13	0.9349	17.78
E4-B nfc=64	34.09	0.9729	18.27

References

- [1] Yochai Blau and Tomer Michael. NeRFCodec:the perception-distortion tradeoff. In *Proceedings of the IEEE/CVF Conference on Computer, Vision and Pattern Recognition (CVPR)*, pages 1–10, 2018.
- [2] Caihong Li Mary Jane Samonte, Huie Zhang and Haijie Shen. Grand challenges in super-resolution image reconstruction. 2023.
- [3] Tamar Rott Shaham, Tali Dekel, and Tomer Michaeli. Singan: Learning a generative model from a single natural image. In *Proceedings of the IEEE International Conference on Computer Vision*, 2019.